

DockLight architecture

A concise reference for extending the dock without mixing layout policy, platform surface mechanics, application-window control or autohide behavior.

ARCHITECTURAL
RULE

The layout engine decides where. The selected surface backend decides how that geometry is expressed on the active desktop stack. Autohide and application-window control remain separate concerns.

1. Backend design



Selection is centralized in `create_dock_surface_backend()`. `DockWindow` applies generic visual/orientation state, delegates placement, and does not contain platform-selection branches.

Runtime	Selected surface path	What it owns
Plasma · native Wayland	PlasmaWaylandDockSurfaceBackend	gtk-layer-shell setup; monitor, anchors, margins, size and exclusive zone
X11 / XWayland	LegacyDockSurfaceBackend	absolute placement, X11 work-area cache, EWMH struts and legacy surface behavior
GNOME integration	Legacy placement + Shell-side integration where required	compositor actor visibility/effects; it complements surface placement and does not replace layout

SURFACE BACKEND OWNS

- Applying an already-calculated dock rectangle.
- Platform-native anchors, coordinates, work-area reservation and exclusive zones/struts.
- Platform-specific map/unmap or surface rules required by the compositor.

IT MUST NOT OWN

- Dock, item, tooltip or preview geometry calculations.
- Autohide policy, pointer intent, delays or effect selection.
- Application-window discovery/control; that remains behind `WindowBackend`.

Adjacent boundaries

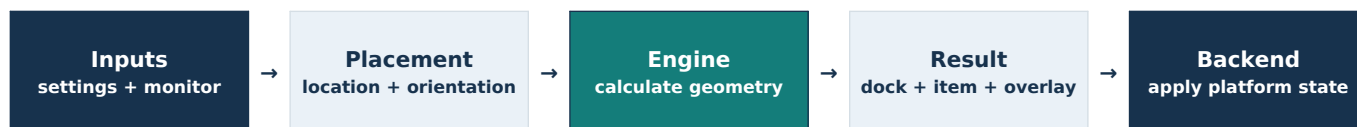
- `WindowBackend` owns discovery, activation, minimization/closing and thumbnail-related access to application windows. It is independent from dock-surface placement.
- Autohide controller owns policy, selected effect, delays, pointer state, timers and transition reversal. A backend only performs the native surface operation requested by that controller.
- Overlay surfaces (reveal, tooltip and Preview) use calculated geometry and independent lifecycles. They must never be mistaken for the main dock by compositor selectors.

IMPLEMENTED
BASELINE

Surface-backend refactor Steps 1-3 are the reference point: `IDockSurfaceBackend`, platform factory, Plasma Wayland backend and legacy X11/XWayland backend. Last recorded automated baseline after fade work: 22/22 tests passed.

2. Layout architecture

Geometry is a deterministic data flow. Desktop-specific APIs are applied only after the dock, item and overlay rectangles are known.



Component	Role	Contract
DockLayoutConfig	Static geometry policy	dock margin/height; tooltip size/distance; hover scale
DockPlacement	Normalized intent	location, alignment, anchors, requested size and derived orientation
DockLayoutEngine	Runtime calculation	monitor + placement + item state → deterministic geometry
Geometry structs	Data contract	MonitorGeometry, DockWindowGeometry and ItemGeometry
DockWindow	Orchestration	collect inputs, ask engine to calculate, ask backend to apply

Four-edge axis mapping

Edge	Orientation	Primary axis	Overlay direction	Reservation edge
BOTTOM	horizontal	X	above dock	bottom
TOP	horizontal	X	below dock	top
LEFT	vertical	Y	right of dock	left
RIGHT	vertical	Y	left of dock	right

The same calculation path covers BOTTOM, TOP, LEFT and RIGHT. Horizontal docks advance items on X; vertical docks advance on Y. Edge-specific code should only map orientation, direction and reservation edge.

GEOMETRY INVARIANTS

- Monitor origin is part of every screen-space calculation; never assume (0,0).
- Center alignment is relative to the selected monitor, not the global desktop.
- Tooltip/preview distance is measured from the visible dock edge and then clamped to monitor bounds.
- One engine path handles all four edges; orientation changes axis mapping, not ownership.

HAND-OFF INVARIANTS

- The engine returns geometry; the backend does not reinterpret or recalculate it.
- Exclusive zones and X11 struts derive from the final dock geometry.
- Work-area and compositor quirks remain inside the selected backend.
- Overlay windows stay independent surfaces with their own lifecycle and future identity.

Runtime calculation contract

- Capture the selected monitor's origin and dimensions, current placement, content/item sizes and the configuration constants.
- Derive orientation and anchors from location; calculate dock size and centered/start/end/fill alignment without consulting platform APIs.
- Calculate each item center, then position tooltip/Preview from the relevant dock edge and clamp the overlay to the monitor rectangle.
- Return geometry structs. DockWindow forwards the final dock geometry to IDockSurfaceBackend and uses overlay geometry for the corresponding surface.

REVIEW CHECK

If a layout change requires testing for KDE versus GNOME inside DockLayoutEngine, the boundary is probably wrong. If a backend change recomputes centers or distances, the boundary is definitely wrong.

3. Testing strategy & development principles

Automated tests protect deterministic rules; a compact runtime matrix protects compositor behavior that cannot be proven by geometry tests alone.

1 Build Debug + Release	2 Automated unit + regression	3 Architecture boundary assertions	4 Runtime 4 DE/session paths	5 Commit evidence recorded
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Required runtime smoke matrix

Environment	Edges	Placement / reserve	Overlay	Autohide	App windows
Plasma Wayland	B T L R	anchors, margins, work area	tooltip + preview	hide/show + reversal	discover, activate, close
Plasma X11	B T L R	coordinates, struts, top panel	tooltip + preview	slide/fade path	discover, activate, close
GNOME Wayland / XWayland	B T L R	screen + panel offsets	no move flicker	Shell actor path	discover, activate, close
GNOME X11 native	B T L R	legacy coordinates + struts	distance + clamping	legacy effect path	discover, activate, close

For every environment, verify initial placement and monitor selection at all four edges, then repeat with a top panel/work-area offset. Exercise tooltip, multi-card Preview, app activation/close and autohide reversal before declaring the path green.

MANDATORY REGRESSIONS

- Move pointer between icons: tooltip and Preview must not double-show, flash or jump.
- Close the last Preview card: Preview lifecycle must follow explicit pointer/window state.
- Reverse autohide mid-transition on horizontal and vertical docks; no stale timer or flicker.
- Play animated Firefox/YouTube/PiP content and verify live thumbnails where the backend supports it.

CURRENT HARDENING TARGET

- Give dock, reveal, tooltip and Preview distinct surface identities.
- Make KWin selectors match only the intended role, including auxiliary-first startup.
- Keep this identity work separate from layout and WindowBackend refactors.
- Status: planned/ongoing protection; not part of the committed Steps 1–3 baseline documented here.

Development principles

Change one responsibility	Fix the named defect, validate it, then stop. Avoid opportunistic architecture work.
Preserve the boundary	Platform selection and native surface operations stay out of DockWindow and the layout engine.
Prefer deterministic geometry	Inputs and outputs should be inspectable; effects must not alter the placement contract.
Code for maintenance	Use focused GTKmm includes, m_ member names, and comments that explain why—not each line.
Commit evidence	Record build/tests plus the exact platform, session and four-edge smoke result.

DEFINITION OF DONE

The requested defect is fixed without moving responsibility across layers; Debug and Release builds pass; automated tests pass; affected environments pass the four-edge smoke test; relevant regressions are exercised; and the result is committed as a small, reviewable checkpoint.

Suggested commit evidence

Build: Debug ✓ · Release ✓ | Tests: N/N ✓ | Runtime: [DE/session] B/T/L/R ✓ | Regressions: [named cases] ✓